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OH and C_2H_4 species time-histories during hexadecane and diesel ignition behind reflected shock waves

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Abstract

The first simultaneous multi-species laser absorption time-history measurements for OH and C_2H_4 were acquired during the oxidation of *n*-hexadecane and commercial diesel fuel (DF-2). The experiments were performed behind reflected shock waves in a new second-generation aerosol shock tube over a temperature range of 1120K to 1373 K and a pressure range of 4–7 atm. Initial fuel concentrations varied between 150 and 1800 ppm with equivalence ratios between 0.4 and 2, and were determined using 3.39 μm He–Ne laser absorption. OH concentration time-histories were measured using absorption of frequency-doubled ring-dye laser radiation near 306.7 nm. Ethylene time-histories were measured using absorption of CO_2 gas-laser radiation near 10.5 μm. Comparisons are given of these species concentration time-histories with two current large *n*-alkane mechanisms: the LLNL-C-16 mechanism of Westbrook et al. [13] and the Jet-SurF C-12 mechanism of Sirjean et al. [14]. Fair agreement between model and experiment is seen in the peak ethylene yields for both fuels; however, modeled early time-histories of OH, an important chain-branching species, differ significantly from current measurements.
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Keywords: Diesel; Hexadecane; Time-histories; C_2H_4; OH

1. Introduction

Detailed species time-history measurements during reaction of gas-phase diesel fuel and diesel fuel surrogates are needed to test and refine the large reaction mechanisms used to model the ignition of these fuels. Such measurements are difficult to achieve. Gas-phase studies of these fuels in conventional shock tubes typically require extreme heating of both the shock tube and mixing assembly, because of the low fuel vapor pressure. As well, in the case of multi-component fuels such as diesel, normal loading techniques that use the vapor-pressure above the liquid fuel tend to fractionally distill the fuel, and again extreme heating is necessary to avoid these effects [1,2]. High-quality gas-phase experiments are, however, necessary in order to be able to separate the physical processes, such as evaporation, that occur during burning, from the chemical processes, such as oxidation, which are described by the large reaction mechanisms. An alternative strategy to study the gas-phase chemistry of low-vapor-pressure fuels is thus required.

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The recent development of an aerosol shock tube [3,4] provides a new approach for gas-phase studies of low-vapor-pressure fuels. Using this method, low fuel loading and fractional distillation can be avoided, as the fuel is introduced into the shock tube at room temperature as a spatially-uniform aerosol. This fuel aerosol is then completely evaporated into the gas-phase by the intermediate high temperatures (500–700 K) that momentarily exist behind the incident-shock wave. Gas-phase low-vapor-pressure fuels can then be studied at high temperatures (1000–1500 K) and pressures behind the subsequent reflected shock wave. Using this aerosol shock tube method, we have previously measured ignition delay times in gas-phase *n*-dodecane and diesel [3–5]. Critical to the temperature and mixture accuracy of these measurements is the requirement that the initial test gas mixture be spatially uniform in the shock tube. The second-generation aerosol shock tube used in this present study has been demonstrated to provide even higher spatial uniformity than our first-generation facility [4].

In our laboratory, we have developed sensitive, species-specific, and quantitative UV and visible laser absorption diagnostics for an array of important combustion species including OH, CH_3, CH, NH_2, and NO [6]. Recently, we have extended our suite of laser absorption diagnostics into the infrared. Using commercially available IR gas and diode lasers we have developed quantitative measurement capabilities for C_2H_4, H_2O, CO_2, CH_4 and selected *n*-alkanes [7–9]. Use of these laser absorption diagnostics in shock tubes has allowed the generation of multi-species time-histories under near-constant-volume conditions intended as targets for chemical kinetic modeling.

To elucidate the inner workings of the large reaction mechanisms describing diesel and diesel-surrogate ignition chemistry, we have applied this multi-species strategy to two important transient species that appear during diesel and *n*-hexadecane oxidation: OH, and C_2H_4. Time-history measurements of OH are extremely useful in quantifying ignition processes, as they provide critical information about the radical-pool population. OH profiles in these systems typically show an immediate rapid rise (that is dependent on fuel decomposition kinetics) to a relatively long-lived pre-ignition plateau (that is dependent on the slower oxidation kinetics of the stable alkene intermediate species), and a final strong exponential rise during ignition. Time-history measurements of ethylene, C_2H_4, provide quantitative information on the main high-temperature decomposition product and pathways of alkanes. Other groups have examined multiple species during hexadecane and diesel combustion in a jet-stirred reactor (JSR) [10–12], but these studies do not provide the detailed information on the short, transient time scales available in shock tubes.

Here we present measurements of OH and C_2H_4 during the oxidation of diesel fuel (DF-2) and a single-component large-alkane surrogate for diesel fuel, *n*-hexadecane. Comparisons are given of these species concentration time-histories with two current large *n*-alkane mechanisms: the LLNL-C-16 mechanism of Westbrook et al. [13] and the JetSurF C-12 mechanism of Sirjean et al. [14].

2. Experimental setup

2.1. Aerosol shock tube

The experiments were performed in a new second-generation aerosol shock tube (AST II). This facility is shown schematically in Fig. 1. The shock tube consists of a driver and driven section. At the downstream end of the driven section is a 1.2 m long square test section (10 × 10 cm) that is contained between two gate valves. This is the portion of the shock tube that is initially filled with aerosol. Just before the incident-shock wave is initiated, the gate valves are set to the configuration shown in Fig. 1 where the driven gate valve is open and the endwall gate valve is closed. As the incident-shock wave travels through the aerosol, the small droplets are quickly evaporated, leaving behind a purely gas-phase mixture. Thus behind the reflected shock the test gas mixture can be considered fully evaporated and fully mixed. The measured spatial variations in concentrations of the resulting mixture are typically less than 1% [4]. Species concentration time-history measurements are then made using laser absorption behind the reflected shock wave through sidewall windows located at 4 cm from the endwall. In Fig. 1, the reflected shock has propagated nearly all the way through the aerosol mixture (labeled C).

2.2. Laser absorption

Three CW laser systems are used in these experiments: an IR He–Ne laser to measure initial

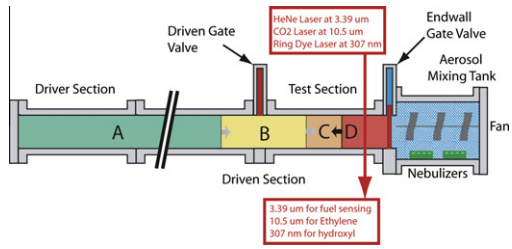

Fig. 1. Schematic of aerosol shock tube (AST II) setup. The different regions in the tube are (from left to right): (A) driver gas, (B) incident-shock heated driven gas with no fuel, (C) incident-shock heated test mixture with fuel, and (D) reflected shock heated test mixture.

fuel concentration; an IR CO_2 gas-laser to measure ethylene; and a UV ring-dye laser to measure OH concentration.

Initial fuel concentration measurements were by laser absorption at 3.391 μm using a low-noise IR Helium–Neon laser (Jodon model HN-10 G-IR). Significant interference absorption by fuel decomposition products that are formed during high-temperature oxidation prevent this single-wavelength absorption signal from being used to monitor the full time-histories of the parent fuel.

Ethylene was measured at two wavelengths using tunable CO_2 laser absorption. The wavelengths that were used for this experiment were 10.532 μm for "on-line" C_2H_4 measurements (strongest absorption by C_2H_4) and 10.675 μm for "off-line" C_2H_4 measurements (weak absorption by C_2H_4). The laser was tuned to the specific CO_2 transition (P(14) for on-line and P(28) for off-line) Potential close-lying transitions of the CO_2 laser were suppressed by first passing the laser beam through a monochromator.

A common-mode rejection scheme was used to reduce laser noise (for all three laser systems), employing both reference (I_0) and transmitted (I) beams. The transmitted CO_2 laser beam passed through the shock tube, through a narrow band-pass filter, and was focused onto a LN_2 cooled MCT detector (Fermionics PV-12-100). The detection scheme is similar to that used by Pilla et al. [7]. Species concentration for ethylene (and for fuel and OH) were determined using the Beer–Lambert relation: $I/I_0 = \exp(-k_\lambda PXL)$, where k_λ is the absorption coefficient [$atm^{-1}\,cm^{-1}$], P is the total pressure [atm], X is the absorbing species mole fraction, and L is the pathlength, 10 cm in the current experiments. Separate reflected shock tube and FTIR experiments, to determine the absorption coefficients for ethylene [7] and fuel, were also performed. Absorption coefficients for OH are well-established [15].

Two wavelengths were used to separate the C_2H_4 absorption from the interfering absorption of other alkene species formed during ignition. Near 10.5 μm, the secondary (interfering) absorbers are primarily larger olefins, such as C_3H_6, C_4H_8, and the combustion products H_2O and CO_2 (significant absorption from the combustion products occurs only after ignition, when ethylene concentrations are negligible, and is not included in the following analysis). The wavelengths of the absorption peaks of these olefins are significantly removed from, and do not overlap with, the C_2H_4 features at 10.532 and 10.675 μm, the wavelengths of the P(14) and P(28) transitions, respectively. At these two wavelengths, absorption from the far wings of these olefins bands is assumed to be approximately constant with wavelength. With this assumption and accurate measurements of the absorption coefficient of C_2H_4 at the two wavelengths, we are able to separate the absorption contributions of the ethylene and interfering absorbers, and uniquely determine the C_2H_4 concentration time-history.

OH concentration time-history was measured using laser absorption of the $R_1(5)$ transition in the OH $A^2\sum^+ - X^2\pi(0,0)$ band at 306.7 nm. This UV wavelength was generated using a 532 nm laser (Coherent Verdi) to pump a ring-dye laser (Spectra Physics 380) that was intra-cavity frequency-doubled using a temperature-tuned AD*A crystal. The laser set up was similar to that used by Herbon et al. [15] and others in our laboratory.

A similar two-wavelength technique was used to separate the OH absorption from interference absorption seen near 306 nm. In this wavelength region there is broad-band absorption, likely from conjugated olefins (e.g. 1,3-butadiene). Off-line measurements (32610.4 cm^{-1}) between the $R_1(5)$ and the $R_1(7)$ transitions have effectively no absorption by OH, and any interference absorption (assumed to be constant with small wavelength change) can be directly subtracted from the on-line measurements to give a direct measure of the OH concentration time-history. As in the case of ethylene, two shock wave experiments are required at near-identical conditions to acquire the measurements at two wavelengths and effect the subtraction. Representative data using this method are shown in Fig. 2. High SNR using the UV OH laser absorption diagnostic enables accurate subtraction of the two signals and results in minimum detection limits for OH of 1–10 ppm.

Strong interference absorption in the UV near 306 nm was seen in all of the diesel experiments as indicated in Fig. 2. However, the hexadecane experiments showed no significant UV interference absorption. Related behavior was seen by

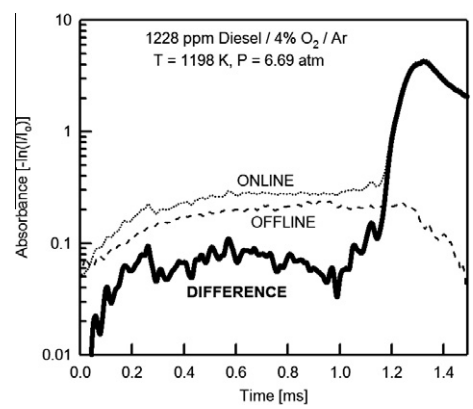

Fig. 2. Diesel OH on-line and off-line laser absorption near 306.7 nm and the difference absorbance signal attributable solely to OH during diesel oxidation. The pre-ignition plateau corresponds to around 10 ppm of OH. Initial reflected shock conditions: 1198 K, 6.69 atm, 1228 ppm diesel, 4% O_2 in argon.

Davidson et al. [16], where interference absorption was observed during iso-octane oxidation, but not in measurements during oxidation of normal alkanes [17].

3. Results and discussion

Presented here are ignition delay time data and laser absorption measurements of OH and C_2H_4 concentration time-histories during the high-temperature oxidation of *n*-hexadecane and diesel fuel. These measurements are also compared with model simulations of *n*-hexadecane oxidation using the LLNL-C16/Westbrook et al. mechanism [13], and model simulations of *n*-dodecane and *n*-heptane oxidation using the JetSurF 1.0/Sirjean et al. mechanism [14]. All calculations were performed using the Reaction Design Aurora suite of CHEMKIN-based codes and implemented with a constant UV (constant internal energy – constant-volume) constraint.

n-Hexadecane was procured from Sigma–Aldrich (99+%) and degassed only before use. The diesel sample (DF-2) was procured locally and degassed only before use. Analysis by IAC Laboratories San Francisco indicate that for the DF-2 fuel, the C/H mass ratio is 6.26, the volume ratio of aromatic:olefin:saturated is 16.2:2.7:81.0, and the cetane index is 45.7. The chemical formula used for equivalence ratio determinations for the diesel fuel was $C_{12}H_{23}$.

OH and C_2H_4 concentration time-histories presented in this paper are given in the accompanying Supplementary Data files.

3.1. Ignition delay times

Figure 3 shows a summary ignition delay time plot of the measurements in the current study: the four *n*-hexadecane 1% O_2, $\phi = 1$ experiments and the ten diesel 4% O_2, $\phi = 0.6$ experiments. Also shown are previous diesel ignition delay time measurements with 21% O_2, $\phi = 0.5$ for a diesel sample with a similar cetane index (CI = 45) [5]. The ignition delay time was defined as the time interval between the arrival of the reflected shock at the sidewall observation (location 4 cm from the end wall) and the extrapolation of the maximum slope of the pressure trace to the zero signal level at the onset of ignition. Similar activation energies are seen in both diesel data sets ($E_A = 32.1$ kcal/mol), while a slightly weaker activation energy is seen in the small *n*-hexadecane data set ($E_A \sim 31$ kcal/mol).

Uncertainties in the scaled ignition delay time measurements are ±15%. The major contributor to this uncertainty is from the temperature; smaller uncertainties can be attributed to the pressure and equivalence ratio scaling used.

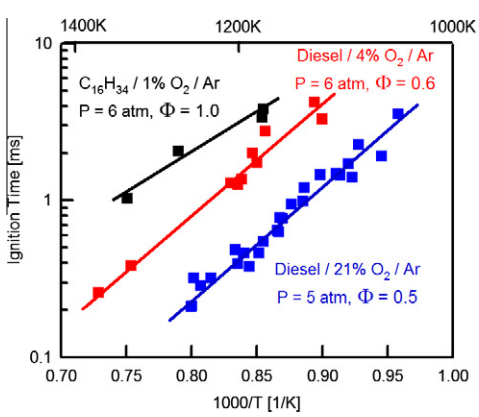

Fig. 3. Ignition delay times for diesel and *n*-hexadecane. All data normalized to common conditions using P^{-1} and ϕ^1 scaling.

3.2. n-Hexadecane oxidation

Figure 4 shows OH and C_2H_4 time-histories during *n*-hexadecane oxidation at 1267 K, 6.54 atm, in 1% O_2/argon with an equivalence ratio $\phi = 1.21$. Excellent signal to noise ratio (SNR) is seen at all times in the C_2H_4 data; however, the low levels of OH measured (~2 ppm) before ignition result in a smaller SNR. The simulations shown using the Westbrook et al. [13] C-16 mechanism (labeled LLNL model) effectively capture the formation rate and nearly capture the pre-ignition C_2H_4 plateau value. Similarly, the modeled time of ignition, indicated by the rapid rise in OH concentration matches the measurement, though the measured OH pre-ignition plateau (at times of 0–1000 μs) is approximately twice the modeled value.

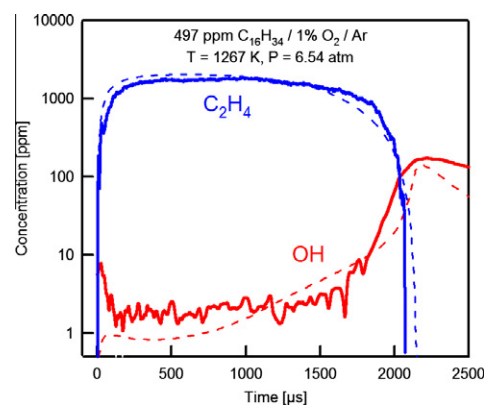

Fig. 4. OH and C_2H_4 species time-histories during *n*-hexadecane oxidation. Initial reflected shock wave conditions: $\phi = 1.2$, 1267 K, 6.54 atm; initial test gas mixture: 497 ppm $C_{16}H_{34}$, 1% O_2/argon. The simulation is based on a LLNL hexadecane mechanism by Westbrook et al. [13].

Figures 5 and 6 show how the C_2H_4 and OH time-histories and the Westbrook et al. C-16 mechanism simulations vary with temperature. The simulations for the lowest and highest temperature examples overpredict the ethylene yields, while the middle temperature (1170 K) simulation is only slightly overpredicted. From the overpredictions of the ethylene yields, it can be inferred that fuel decomposition product yields of other larger olefins (e.g. propene and butene) are likely underpredicted. As well, at these high temperatures, formation times of C_2H_4 are directly related to fuel decomposition rates and pathways, and future efforts to match these profiles should result in improved rates for the decomposition and oxidation pathways for this large *n*-alkane.

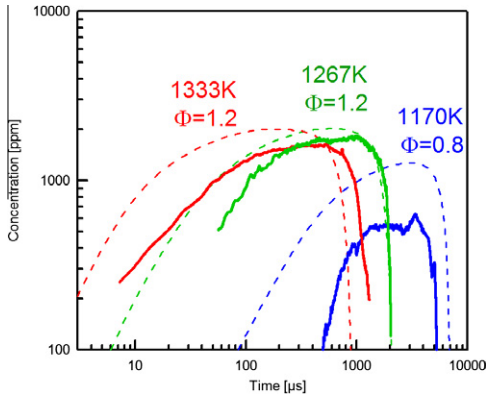

Fig. 5. C_2H_4 species time-histories during *n*-hexadecane oxidation. Initial reflected shock wave conditions: $\phi = 0.8$, 1170 K, 4.60 atm, 326 ppm $C_{16}H_{34}$, 1% O_2/argon; $\phi = 1.2$, 1267 K, 6.54 atm, 497 ppm $C_{16}H_{34}$, 1% O_2/argon; and $\phi = 1.2$, 1333 K, 6.77 atm; 493 ppm $C_{16}H_{34}$, 1% O_2/argon. Modeled using the LLNL model by Westbrook et al. [13].

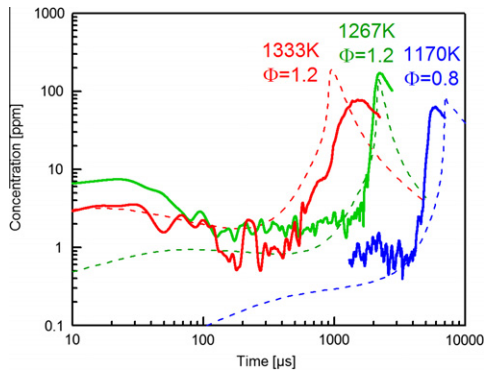

Fig. 6. OH species time-histories during *n*-hexadecane oxidation. Initial reflected shock wave conditions: $\phi = 0.8$, 1170 K, 4.60 atm, 326 ppm $C_{16}H_{34}$, 1% O_2/argon; $\phi = 1.2$, 1267 K, 6.54 atm, 497 ppm $C_{16}H_{34}$, 1% O_2/argon; and $\phi = 1.2$, 1333 K, 6.77 atm, 493 ppm $C_{16}H_{34}$, 1% O_2/argon. Modeled using the LLNL model by Westbrook et al. [13].

Peak OH concentrations are accurately modeled at the two lower temperatures (1170 and 1267 K) simulations, but not for the highest temperature case. Given that OH concentrations are closely related to the other important small radical species (H-atoms, O-atoms and HO_2 radicals) that contribute to chain branching, efforts to refine rates and pathways in the mechanism that improve predictions of OH should improve predictions of all the small radicals contributing to ignition progress. A detailed review of all chain-branching reactions would be required to understand the cause of the discrepancy between the model and the measurements.

3.3. Diesel oxidation

Figure 7 shows OH and C_2H_4 time-histories during diesel oxidation at 1198 K, 6.69 atm, in 4% O_2/argon with an equivalence ratio $\phi = 0.54$. As in the case of *n*-hexadecane, good SNR is seen in the ethylene trace. Interestingly, improved SNR is seen at early times in the OH trace. Simulations using three single-component diesel fuel surrogates, C_7H_{16}, $C_{12}H_26$ and $C_{16}H_{34}$, (using mixtures with identical oxygen mole fraction and equivalence ratios) are also shown. While all three of the diesel fuel surrogate simulations give similar ethylene peak yields (and similar OH peak values), none of the simulations closely matches the ignition delay time (derived from the time to half peak OH concentration). As well, none of the three single-component surrogate models accurately predicts early-time OH concentrations; and all three models underestimate the start of

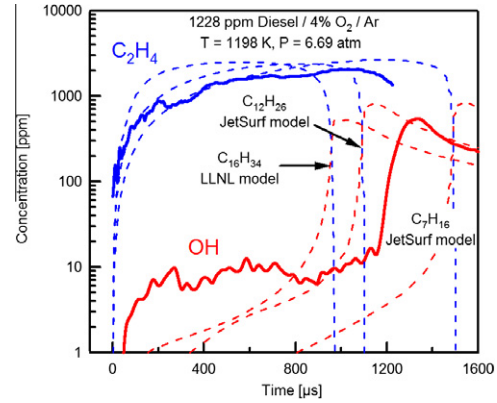

Fig. 7. OH and C_2H_4 species time-histories during diesel oxidation. Initial reflected shock wave conditions: $\phi = 0.5$, 1198 K, 6.69 atm; initial test gas mixture: 1228 ppm diesel, 4% O_2/argon. Simulations shown for single-component surrogate models are: *n*-heptane, and n-dodecane use the JetSurf mechanism and *n*-hexadecane uses the LLNL mechanism.

the OH plateau by an order of magnitude. Given that this diesel sample is 81% saturated paraffins, it might have been expected to be modeled relatively well by a single large *n*-alkane. The approximately ±30% variation between the modeled (JetSurf and LLNL models) and experimental ignition delay times is a measure of the goodness-of-fit one can expect using any of the current single-component diesel surrogates. Post-ignition values of ethylene are expected to be near zero, however absorption by the combustion products H_2O and CO_2 begins to appear at ignition, and thus the absorption-based C_2H_4 profile is not shown beyond 1200 μs.

Figures 8 and 9 present variations in the OH and C_2H_4time-history measurements as a function of temperature for diesel oxidation. In both the OH and the C_2H_4time-histories, modeled values of peak yields (using the *n*-hexadecane single-component surrogate) are relatively close to (but consistently exceed) the measured values at all temperatures. Time scales in the simulations for formation of OH and C_2H_4 typically vary by a factor of two from the experiment. Given the current relatively confident state of knowledge about large *n*-alkane pyrolysis and oxidation chemistry (albeit tempered by the comparisons shown earlier for *n*-hexadecane), the differences seen between the single-component surrogate modeling and the diesel experiments indicates that a multi-component surrogate for diesel will be needed to capture the finer details of these experiments. In particular, interfering absorption near 306.7 nm that indicates the presence of UV absorbing species (i.e. conjugated olefins) during diesel oxidation chemistry, but not during *n*-hexadecane oxidation, point to a need to include these larger, more chemically complex, intermediate species in diesel modeling.

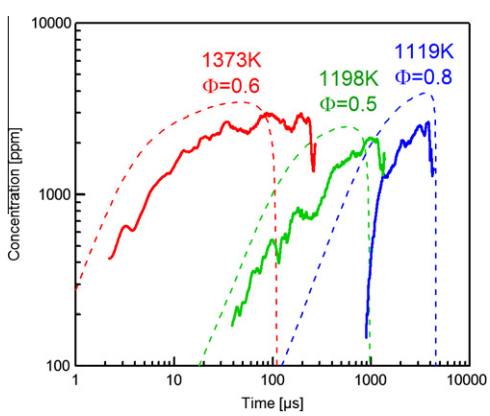

Fig. 8. C_2H_4 species time-histories during diesel oxidation. Initial reflected shock wave conditions: $\phi = 0.8$, 1119 K, 4.63 atm, 1761 ppm diesel, 4% O_2/argon; $\phi = 0.5$, 1198 K, 6.69 atm, 1228 ppm diesel, 4% O_2/argon; and $\phi = 0.6$ 1373 K, 6.33 atm, 1380 ppm diesel, 4% O_2/argon. Modeled using the LLNL model by Westbrook et al. [13] for hexadecane.

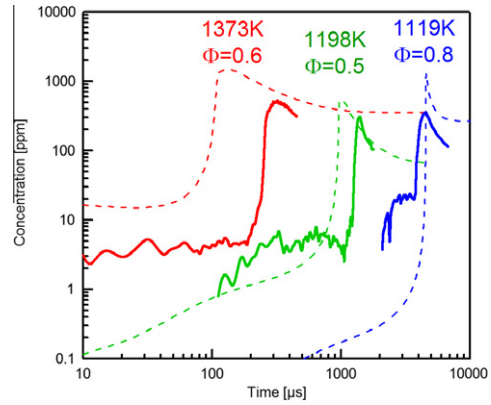

Fig. 9. OH species time-histories during diesel oxidation. Initial reflected shock wave conditions: $\phi = 0.8$, 1119 K, 4.63 atm, 1380 ppm diesel, 4% O_2/argon. Modeled using the LLNL model by Westbrook et al. [13].

The increased noise in the C_2H_4 profiles in Fig. 8 is also a result of a reduction in SNR due to subtraction of absorption from interfering species.

4. Conclusions

The first multi-species time-history measurements for high-temperature *n*-hexadecane and diesel oxidation are presented. Measured time-histories for OH and C_2H_4 are compared with single-component diesel surrogate models. Agreement in ignition delay times and species concentrations typically are within a factor of two at all temperatures. However, early-time predictions of OH mole fraction, indicative of the ability of these mechanisms to capture the important chain-branching species concentrations, vary widely from the experiments.

These measurements provide quantitative kinetic targets for the testing and refinement of *n*-hexadecane and diesel-surrogate models. Further work is planned to extend species time-history measurements for these fuels to include simple alkanes (i.e., methane and ethane), larger alkenes particularly propene and butene, and conjugated alkenes (e.g., 1,3-butadiene).

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Appendix A. Supplementary data

Supplementary_Material_Haylett.xls – Contains ignition delay time data and conditions. Also

contains all species time-history data found in this paper. Supplementary data associated with this article can be found, in the on-line version, at doi:10.1016/j.proci.2010.05.053.

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